

Spring 2000

Problem 1 (Sp00). Are the 4×4 matrices

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -1 & 0 & 0 & 0 \\ -1 & 1 & 1 & -1 \\ -1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{pmatrix}$$

similar?

Problem 2 (Sp00, Fa23). Let $\{f_n\}$ be a uniformly bounded, pointwise equicontinuous sequence of real valued functions on the compact metric space (X, d) . Define the functions $g_n : X \rightarrow \mathbb{R}$, for $n \in \mathbb{N}$ by

$$g_n(x) = \max\{f_1(x), \dots, f_n(x)\}$$

Prove that the sequence $\{g_n\}$ converges uniformly.

Problem 3 (Sp00). Prove that the group $G = \mathbb{Q}/\mathbb{Z}$ has no proper subgroup of finite index.

Problem 4 (Sp89, Sp00). Let f and g be entire functions such that

$$\lim_{z \rightarrow \infty} f(g(z)) = \infty$$

Prove that f and g are polynomials.

Problem 5 (Fa87, Sp92, Fa97, Sp00, Fa13, Sp19). Define a sequence of real numbers (x_n) by

$$x_0 = 1 \quad x_{n+1} = \frac{1}{2 + x_n} \quad \text{for } n \geq 0$$

Show that (x_n) converges and evaluate its limit.

Problem 6 (Sp00). Let A be an $n \times n$ matrix over $\mathbb{C}$ whose minimal polynomial μ has degree k .

1. Prove that, if the point λ of $\mathbb{C}$ is not an eigenvalue of A , then there is a polynomial p_λ of degree $k - 1$ such that $p_\lambda(A) = (A - \lambda I)^{-1}$.

2. Let $\lambda_1, \dots, \lambda_k$ be distinct points of $\mathbb{C}$ that are not eigenvalues of A . Prove that there are complex numbers $c_1, \dots, c_k$ such that

$$\sum_{j=1}^k c_j (A - \lambda_j I)^{-1} = I$$

Problem 7 (Sp00). Let f be a positive function of class $\mathcal{C}^2$ on $(0, \infty)$ such that $f' \leq 0$ and f'' is bounded. Prove that $\lim_{t \rightarrow \infty} f'(t) = 0$.

Problem 8 (Sp00). Find the cardinality of the set of all subrings of $\mathbb{Q}$.

Problem 9 (Sp00). Evaluate $I = \int_{|z|=1} \frac{\cos^3 z}{z^3} dz$ where the direction of integration is counterclockwise.

Problem 10 (Sp00). Let S be an uncountable subset of $\mathbb{R}$. Prove that there exists a real number t such that both sets $S \cap (-\infty, t)$ and $S \cap (t, \infty)$ are uncountable.

Problem 11 (Sp00). Let A_n be the $n \times n$ matrix whose entries a_{ij} are given by

$$a_{ij} = \begin{cases} 1 & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

Prove that the eigenvalues of A_n are symmetric with respect to the origin.

Problem 12 (Sp00). Suppose that H_1 and H_2 are distinct subgroups of a group G such that $[G : H_1] = [G : H_2] = 3$. What are the possible values of $[G : H_1 \cap H_2]$?

Problem 13 (Sp00). Let f be a nonconstant entire function whose values on the real axis are real and nonnegative. Prove that all real zeros of f have even order.

Problem 14 (Sp00). Let $I_1, \dots, I_n$ be disjoint closed nonempty subintervals of $\mathbb{R}$.

1. Prove that if p is a real polynomial of degree less than n such that

$$\int_{I_j} p(x) dx = 0 \quad \text{for } j = 1, \dots, n$$

then $p = 0$.

2. Prove that there is a nonzero real polynomial p of degree n that satisfies all the above equations.

Problem 15 (Sp00). Let F , with components $F_1, \dots, F_n$, be a differentiable map of $\mathbb{R}^n$ into $\mathbb{R}^n$ such that $F(0) = 0$. Assume that

$$\sum_{j,k=1}^n \left| \frac{\partial F_j(0)}{\partial x_k} \right|^2 = c < 1$$

Prove that there is a ball B in $\mathbb{R}^n$ with center 0 such that $F(B) \subset B$.

Problem 16 (Sp00). Let A be a complex $n \times n$ matrix such that the sequence $(A^n)_{n=1}^\infty$ converges to a matrix B . Prove that B is similar to a diagonal matrix with zeros and ones along the main diagonal.

Problem 17 (Sp00). Evaluate the integrals $I(t) = \int_{-\infty}^{\infty} \frac{e^{itx}}{(x+i)^2} dx$ for values $-\infty < t < \infty$.

Problem 18 (Sp00). Let G be a finite group and p a prime number. Suppose a and b are elements of G of order p such that b is not in the subgroup generated by a . Prove that G contains at least $p^2 - 1$ elements of order p .